

ONLINE LIBRARY MANAGEMENT

1. Abstract

The Online Library Management System is a web-based application designed to manage library activities efficiently and systematically. It helps to maintain information about books, students, authors, book availability, issue details, and return records in a centralized database.

The system allows the librarian to add, update, delete, and search book records. Students can search for available books and view their borrowing details. The system also helps to track issued and returned books and reduces manual paperwork.

The main objective of this project is to provide a simple, secure, and user-friendly system that saves time, reduces errors, and improves the overall efficiency of library management.

2. Methodology

The Online Library Management System is developed using a systematic software development process.

1. Requirement Analysis:

The requirements of the library are identified, such as book management, student management, issue and return management, and book searching.

2. System Design:

The system structure, user interface, database, and different modules are designed according to the requirements.

3. Database Design:

A database is created to store book details, student details, issue records, return records, and other library information.

4. Development:

The front-end and back-end components are developed and connected with the database to provide the required library functions.

5. Testing:

Each module is tested to identify and correct errors and to ensure that the system works properly.

6. Implementation:

After successful testing, the system is implemented for use by the librarian and students.

7. Maintenance:

The system is regularly updated and maintained to improve security, performance, and functionality.

3. Conclusion

The Online Library Management System provides an efficient and convenient way to manage library operations. It reduces manual work and helps maintain book and user information accurately in a centralized database.

The system makes it easier to search books, manage book availability, issue and return books, and maintain library records. It saves time and improves the efficiency of both librarians and users.

Thus, the Online Library Management System is a useful solution for modern libraries. In the future, additional features such as online book reservation, email notifications, digital books, and automated reports can be added to improve the system further.